

Build a Web AI App

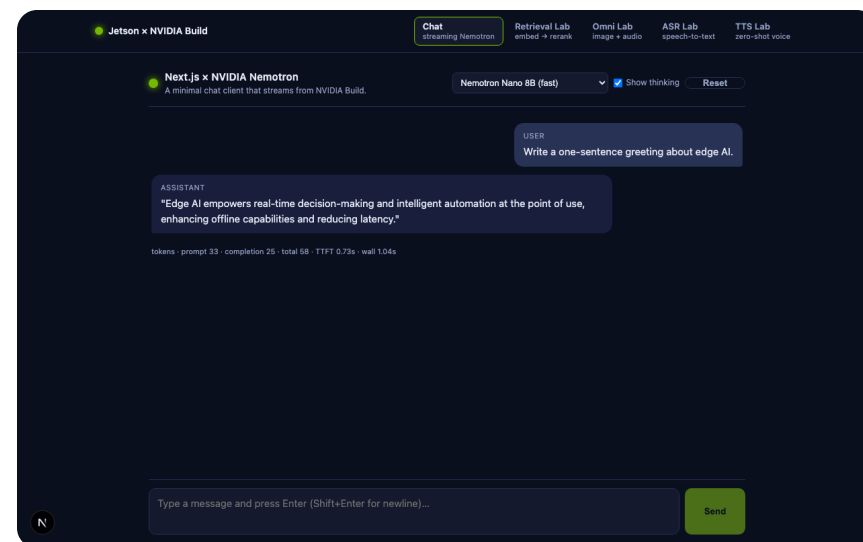
Next.js + NVIDIA Nemotron (Build API)

SJSU · Edge AI

A streaming chatbot web app that runs on your Jetson and serves to any browser.

1 What you'll build

- A **Next.js** web app with a **streaming chat** UI.
- Talks to **NVIDIA Nemotron** (and optionally **OpenAI** / **Anthropic**) via cloud APIs.
- **Runs on the Jetson**, opened from your laptop's browser.
- 🗝️ Your API key stays **server-side** — never sent to the browser.

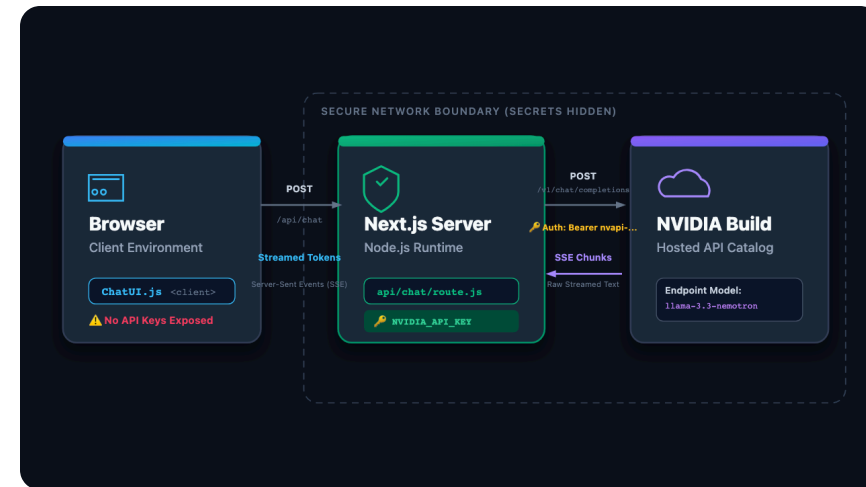


2 The big picture

Your browser talks to a small **server on the Jetson**, which talks to the model API:

Browser → /api/chat (on Jetson) →
NVIDIA/OpenAI/Anthropic → stream back

- The **page** runs in the browser (buttons, live text).
- The **API route** runs on the server — it holds the key and calls the model.
- The browser **never sees the key**.



3 Key web concepts (background)

- **Next.js** = a React framework. Each folder under `app/` is a page (the *App Router*).
- **Server Components / routes** run on the Jetson — they can hold **secrets** and call APIs.
- **Client Components** run in the **browser** — they're interactive (`"use client"`).
- **Streaming (SSE)**: the server forwards tokens one chunk at a time, so the answer appears live.

Rule of thumb: **secrets + external API calls** → **server**; **buttons + live updates** → **client**.

4 Where's the code (and what each part does)

```
edgeLLM/nextjs-nemotron-app/  
  app/page.js           # the chat UI – Client Component (browser)  
  app/components/ChatUI.js # streaming chat box  
  app/api/chat/route.js  # server route – holds the key, streams from the model  
  app/api/models/route.js # the model dropdown list  
  lib/providers.js       # picks NVIDIA/OpenAI/Anthropic + reads keys from ~/.env.local  
  .env.local            # optional local keys (git-ignored)
```

 Repo: [edgeLLM/nextjs-nemotron-app](https://github.com/edgeLLM/nextjs-nemotron-app)

5 Setup — keys + Node

Keys come from your `~/.env.local` (the same file `sjsujetsontool chat` saved). Add any of:

```
echo "NVIDIA_API_KEY=nvapi-..." >> ~/.env.local # build.nvidia.com (free)
echo "OPENAI_API_KEY=sk-..." >> ~/.env.local # platform.openai.com
echo "ANTHROPIC_API_KEY=sk-ant-..." >> ~/.env.local # console.anthropic.com
```

The app picks the provider from the **model you choose** (`nvidia/...` , `gpt-...` , `claude-...`).

- **Node/npm live in the container** — there's no `npm` on the host. Run everything inside `sjsujetsontool shell` (it also passes your `~/.env.local` keys into the container).

6 Run it

```
sjsujetsontool shell          # Node 20 + npm + your keys
cd /Developer/edgeAI/edgeLLM/nextjs-nemotron-app
npm install                   # first time (~30-60 s)
npm run dev                   # serves on 0.0.0.0:3000
```

Open it **from your laptop** using the Jetson's IP:

```
hostname -I | awk '{print $1}'    # e.g. 192.168.5.206 → http://192.168.5.206:3000
```

Each message streams: Jetson → model API → Jetson → your browser, with a TTFT / tokens-per-second line.

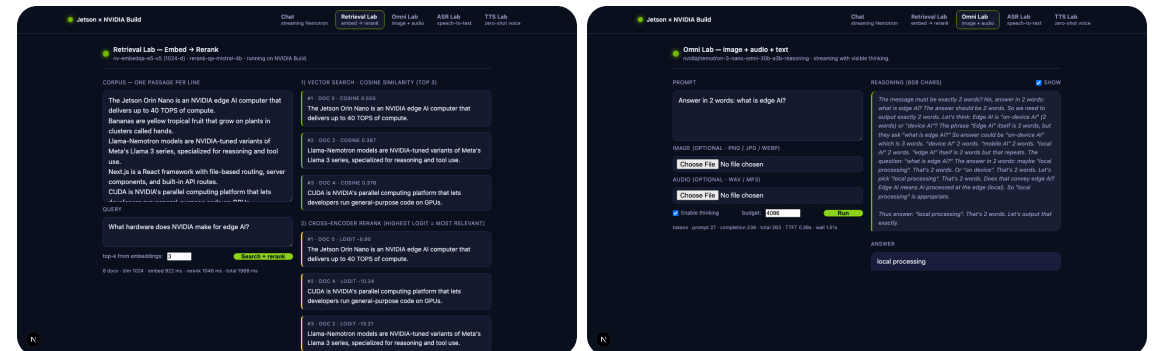
7 Extend it — same pattern every time

Every feature = **one page** (UI) + **one API route** (server logic). Copy the pattern:

```
app/<feature>/page.js      # the page/UI
app/api/<feature>/route.js  # server: read key, call a model, return/stream
```

The bonus labs are exactly this — add a route + page and you've extended the app:

 **Retrieval** ·  **Omni** (vision) ·  **ASR** ·  **TTS**



8 Make it your own (push to GitHub)

Copy the app into your own folder, then create your repo:

```
cp -r /Developer/edgeAI/edgeLLM/nextjs-nemotron-app ~/my-ai-app
cd ~/my-ai-app && rm -rf node_modules .next
git init && git add -A && git commit -m "My Edge AI web app"
```

Create an empty repo on github.com (the + → **New repository**), then:

```
git remote add origin https://github.com/<your-username>/my-ai-app.git
git branch -M main && git push -u origin main
```

With the GitHub CLI it's one line: `gh repo create my-ai-app --public --source=. --push`

Full walkthrough

Step-by-step build, code, and the four bonus labs:

lkk688.github.io/edgeAI/curriculum/11_nextjs_nemotron_app

← Back to the [Get-Started slides](#)